

Task Coordinates for Few-Shot Diffusion Adaptation: Results Against the Experimental Plan

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Abstract

Meta-Diffusion adapts a frozen denoising diffusion probabilistic model to a new data domain by fitting a low-dimensional task coordinate rather than by updating network weights. This report gives the results of the current experimental stage, block by block against the governing plan. The plan fixes the criterion in advance: the main claim holds only if the correct source task together with a learned source-to-target relation outperforms *both* relation-only transfer *and* target-only few-shot modelling. That criterion is met when target data are scarce, and only then. Six of the ten deliverables are complete; four are not, and each is named with its reason.

1 CRITERION AND NOTATION

Criterion. The main claim is supported only if the correct source task *plus* the learned source-to-target relation outperforms *both* relation-only transfer *and* target-only few-shot modelling, in the low-target-data regime.

Notation. A set encoder maps a set of images to a task coordinate $z \in \mathbb{R}^k$. Writing z_S for the coordinate encoded from source-domain data and c for a descriptor of the source-to-target relation, the transport map predicts the target coordinate $\tilde{z}_T = z_S + \Delta_\gamma(z_S, c)$, taking no target image as input. M_S is the number of source images the encoder sees; K_T is the number of target images available at adaptation time. Feature-wise linear modulation, the standard alternative to an additive basis, scales and shifts intermediate feature maps as a function of the coordinate; it is called the modulation arm below.

Metric. Distances are the sliced Wasserstein distance to the real target set in pixel space, which the plan names as headline; lower is better. Every comparison is a paired difference over episodes with a 95% confidence interval, and an asterisk marks an interval excluding zero.

Configuration. One checkpoint of the three-transformation CIFAR-100 domain-shift family — Gaussian blur, additive Gaussian noise, contrast reduction at severity 3 — a U-Net backbone of 128 base channels, coordinate dimension $k = 16$, 50,000 training steps read with exponentially moving averaged weights. Adaptation uses 25 refinement steps at rate 10^{-2} . Generation uses the deterministic implicit sampler, 50 reverse steps, 96 samples per cell, 60 episodes per cell.

2 STAGE 0 — PRECONDITIONS

	Precondition	Status
0.1	Integrate the three branches	Not done. Both modulation-arm source files are absent from this tree. Blocks A2.
0.2	Verify the split file	Done; hash matches the fixed value.
0.3	Modulation checkpoint	Not done. Absent from this machine. Also blocks A2.
0.4	Evaluation primitives	Four of five written. The kernel Inception distance was not implemented.
0.5	Output directory	Done.

Table 1: Stage 0. The five preconditions the plan places before all other work, and whether each holds in the tree that produced the results below.

3 STAGE A — CIFAR-100, GENERATION-LEVEL EVIDENCE

3.1 A1 — Source-dependence controls

180 episodes, twenty classes by three transformations by $K_T \in \{1, 5, 20\}$, nine conditions.

Full method against	$K_T = 1$	$K_T = 5$	$K_T = 20$
Mean z_S	$+0.0301 \pm 0.0072^*$	$+0.0346 \pm 0.0085^*$	$+0.0338 \pm 0.0080^*$
Shuffled z_S	$+0.0416 \pm 0.0125^*$	$+0.0453 \pm 0.0149^*$	$+0.0436 \pm 0.0144^*$
Relation-only	$+0.0254 \pm 0.0078^*$	$+0.0285 \pm 0.0088^*$	$+0.0295 \pm 0.0087^*$
Target-only	$+0.018 \pm 0.009^*$	$+0.011 \pm 0.007^*$	-0.003 ± 0.007
Oracle (negative = behind)	-0.026^*	-0.027^*	-0.017^*

Table 2: A1. Paired differences; positive favours the full method.

Conclusion. The method beats all three source controls at every budget. It beats relation-only transfer and target-only modelling simultaneously at one and five target images, which is what the criterion requires, and the advantage over target-only modelling is gone by twenty. **The criterion is met in the low-target-data regime and is bounded there:** source information is worth having when target evidence is scarce, and its value decays as target evidence accumulates. A gap to the oracle remains at every budget.

3.2 A2 — Additive basis against the modulation arm

Not run. Preconditions 0.1 and 0.3 are both unmet: the modulation implementation is on another branch and its checkpoint is not on this machine. Both are repository operations, not experiments. The arm exists and has been measured on the denoising loss; only the generation-level comparison is outstanding.

3.3 A3 — Target-scarcity sweep

300 episodes at $K_T \in \{1, 2, 5, 10, 20\}$, six strategies.

Conclusion. The method beats reusing the source coordinate at every budget, the advantage decaying monotonically by a factor of five and remaining clear of zero throughout. This is the largest effect in the stage. Against target-only modelling the advantage is confined to the smallest budget. **The plan asked for the crossing point to be named: there is no crossing, only a convergence** — the two become indistinguishable from two target images onward, and the method was never shown to fall behind.

Refinement. Refining the transported coordinate on the available target images changes nothing that can be measured here: its gain over transport alone contains zero at every budget, and so does the remaining gap to the oracle. **Refinement neither helps nor hurts once transport has landed close.** Its value is real elsewhere in the same table — reusing the source coordinate improves from 1.439 to 0.521 across the sweep, and that improvement is refinement

Strategy	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
No adaptation	$1.616 \pm .094$	$1.620 \pm .091$	$1.609 \pm .095$	$1.621 \pm .094$	$1.586 \pm .090$
Reuse z_S	$1.439 \pm .106$	$1.212 \pm .098$	$0.825 \pm .086$	$0.630 \pm .058$	$0.521 \pm .050$
Transport	$0.299 \pm .045$	$0.297 \pm .045$	$0.299 \pm .042$	$0.313 \pm .041$	$0.293 \pm .040$
Transport + refine	$0.291 \pm .045$	$0.294 \pm .047$	$0.295 \pm .036$	$0.312 \pm .043$	$0.322 \pm .048$
Target-only	$0.360 \pm .052$	$0.336 \pm .051$	$0.304 \pm .040$	$0.331 \pm .053$	$0.301 \pm .041$
Oracle	$0.274 \pm .043$	$0.283 \pm .039$	$0.260 \pm .039$	$0.308 \pm .041$	$0.290 \pm .037$

Table 3: A3, absolute distances. All six strategies, in transformation-statistic space, the instrument this sweep was run with; lower is better.

Paired difference	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Transport ahead of reuse z_S <i>half-width</i>	$+1.140^*$ 0.108	$+0.915^*$ 0.105	$+0.526^*$ 0.099	$+0.317^*$ 0.067	$+0.229^*$ 0.059
Transport ahead of target-only <i>half-width</i>	$+0.061^*$ 0.051	$+0.038$ 0.052	$+0.005$ 0.036	$+0.018$ 0.045	$+0.008$ 0.041
Refinement’s gain over transport <i>half-width</i>	$+0.008$ 0.015	$+0.004$ 0.018	$+0.004$ 0.027	$+0.001$ 0.035	-0.029 0.036
Transport + refine behind oracle <i>half-width</i>	$+0.017$ 0.044	$+0.010$ 0.038	$+0.035$ 0.035	$+0.004$ 0.040	$+0.032$ 0.045

Table 4: A3, paired differences. Each row is signed so that a positive value means what the row name says; an asterisk marks an interval excluding zero.

doing the work — which is the pattern to expect from a procedure that closes distance: useful when the starting coordinate is far away, and invisible when it is not. Per the plan, it therefore stays out of the headline method.

3.4 A4 — Source-abundance sweep

Six source budgets from 16 to 300 images, one target image, 60 episodes each.

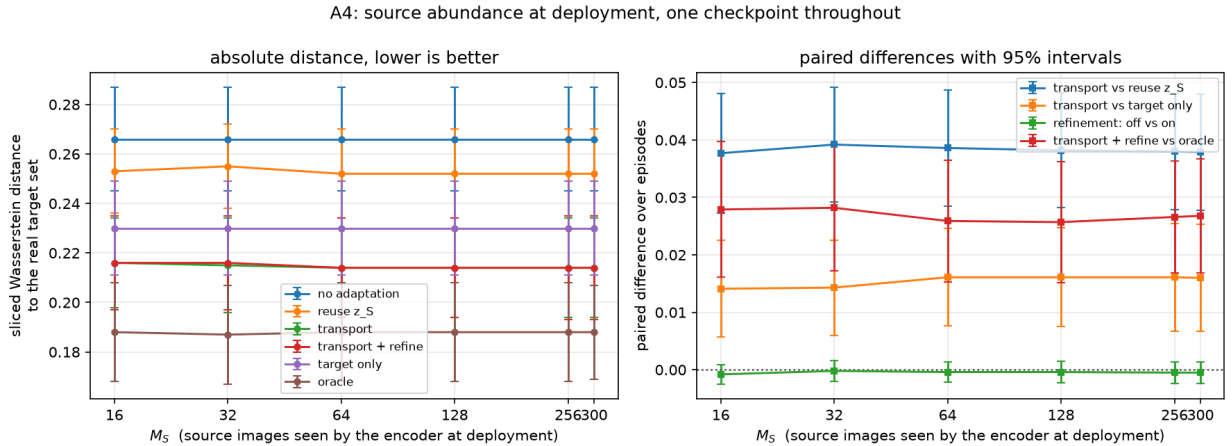


Figure 1: A4. Left: absolute distance per strategy. Right: paired differences with 95% intervals.

Conclusion. Across a nineteenfold range nothing moves: the comparison against reusing the source coordinate shifts by 0.0002, that against target-only modelling by 0.0019, that against the oracle by 0.0011, each smaller than the half-width of its own interval. The plan expected improvement followed by saturation. **What the sweep shows**

is saturation from the first point measured: sixteen source images already buy everything this transport map extracts from a source set. The method is therefore cheap on the source side. The sweep varies source evidence available at deployment; the checkpoint was trained once, at $M_S = 64$.

3.5 A5 — Transport-structure ablation

Four maps trained separately under identical conditions, compared episode by episode over 180 episodes each.

Variant	Parameters	$K_T = 1$	$K_T = 5$	$K_T = 20$
Identity	0	$+0.0242 \pm 0.0080^*$	$+0.0256 \pm 0.0089^*$	$+0.0246 \pm 0.0087^*$
Constant relation vector	96	$+0.0082 \pm 0.0051^*$	$+0.0092 \pm 0.0052^*$	$+0.0109 \pm 0.0055^*$
Linear map	352	$+0.0022 \pm 0.0049$	$+0.0004 \pm 0.0048$	$+0.0036 \pm 0.0050$

Table 5: A5. Each variant paired against the residual multilayer perceptron (6824 parameters) episode by episode; positive favours the residual network. The variants differ only in the form of Δ_γ and are otherwise trained identically.

Conclusion. Both reduced maps are beaten at every budget: the identity by $+0.024$ to $+0.026$ and the constant relation vector by $+0.008$ to $+0.011$, every interval clear of zero. **A transport map is necessary, and a per-relation constant offset is not enough** — the prediction has to depend on the source coordinate itself. The linear map, by contrast, is indistinguishable from the residual network at all three budgets with nineteen times fewer parameters, so **the non-linearity is the part that can be dropped**: the architecture can be simplified rather than defended.

One cell qualifies this. At $K_T = 20$ the constant run is also behind on target-only modelling, by $+0.0135 \pm 0.0081$, a strategy that never consults the transport map at all — so at that budget the difference lies somewhere other than in Δ_γ . At $K_T = 1$ and $K_T = 5$ the same comparison contains zero, and the gap is the map’s.

4 STAGE B — GAUSSIAN-MIXTURE POSITIVE CONTROL

Stage B is run entirely on two-dimensional Gaussian mixtures. It touches no image data. That is its purpose: the mixtures are analytically tractable, so the ground truth is known and the mechanism can be checked against it without the instrument ambiguity that attends any measurement on images. Nothing in this section is evidence about CIFAR-100, and the one figure that shows both is labelled accordingly.

Five task families, 32 tasks each, one separately trained checkpoint per family. The families are ordered by how far apart a task’s source and target distributions are placed: the four labelled by a perturbation size are increasingly distant, and the fifth pairs source and target distributions drawn independently. This axis is the relatedness sweep, included so that task distance can be varied deliberately rather than inferred.

Conclusion. The four-coordinate scan orders itself exactly as the mechanism predicts, in every family: the task’s own coordinate is best, the shared mean and zero are level with each other, and another task’s coordinate is far worse than having none at all. That last point is the informative one. Substituting the wrong task’s coordinate costs roughly 0.23 where removing the coordinate entirely costs nothing, so the coordinate carries task-specific content that actively misleads when it is wrong — which is the signature of a coordinate that is being used rather than absorbed. The profile is minimised at the task’s own coordinate in all five families.

Placed beside the same measurement on images, the well is 26.1% deep on mixtures and 1.6% deep on CIFAR-100, both minimised at the task’s own coordinate. **The mechanism on images is neither absent nor misdirected; it is shallow.** Two observations are unexplained: the relatedness sweep does not move the task-specific term at all although it was built to, and on unrelated tasks the shared mean is *worse* than zero (0.6453 against 0.6149), the only family where that happens.

Task family	perturb 0.1	perturb 0.25	perturb 0.5	perturb 1.0	unrelated
$L(z_y)$, the task’s own	0.4368	0.4750	0.5234	0.5937	0.4921
$L(\bar{z})$, shared mean	0.5907	0.6178	0.6770	0.7508	0.6453
$L(z_{\text{shuffled}})$, another task’s	0.8183	0.8171	0.8634	0.9345	0.7891
$L(0)$	0.5905	0.6178	0.6729	0.7467	0.6149
Task-specific gain, $L(\bar{z}) - L(z_y)$	+0.1539*	+0.1428*	+0.1536*	+0.1571*	+0.1533*
Profile minimum at	$s = 1$	$s = 1$	$s = 1$	$s = 1$	$s = 1$

Table 6: Stage B, the positive control, on **two-dimensional Gaussian mixtures** — not on images. Five checkpoints, one per task family, each $k = 16$ with a linear decoder trained for 80,000 steps and read at $K_T = 5$ over 32 tasks. Columns are the five task families, ordered left to right by increasing distance between a task’s source and target distributions. The first four rows are the denoising loss under four coordinates: the task’s own, the single mean coordinate shared by all tasks, another task’s, and zero. The fifth row is the reduction in denoising loss obtained by giving a task its *own* coordinate rather than the single coordinate shared by all tasks, paired over the 32 tasks; it is the quantity the whole mechanism is supposed to produce, and it should be positive. The second row reports where the coordinate-loss profile of Figure 2 attains its minimum, with $s = 1$ denoting the task’s own coordinate; a minimum anywhere else would mean the coordinate points in the wrong direction, so this row is the correctness check rather than a measurement of size.

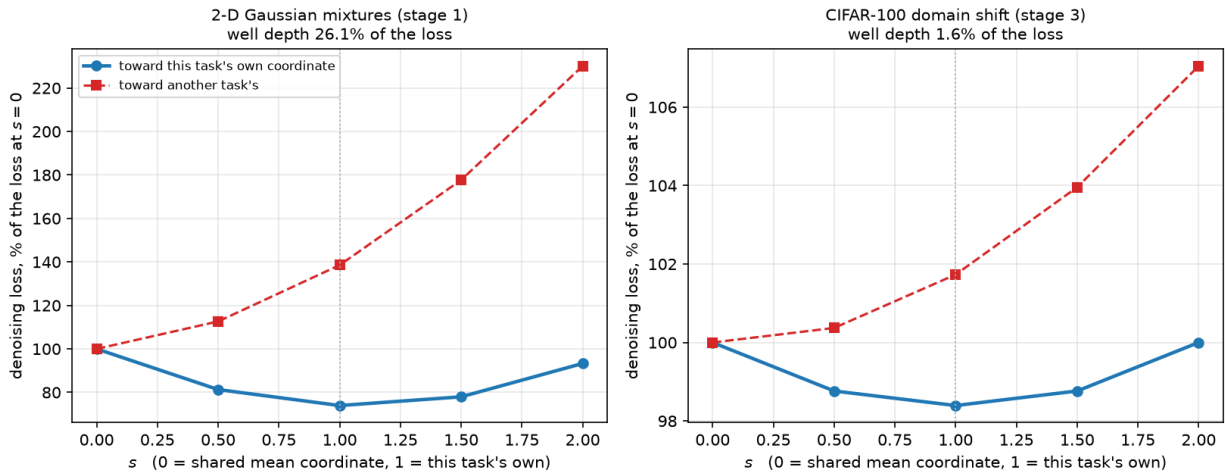


Figure 2: **A cross-stage comparison, not a Stage B measurement.** The left panel is Stage B, on Gaussian mixtures. The right panel is *not*: it is taken from the coordinate controls run on the CIFAR-100 checkpoint in an earlier stage, and is reproduced here only so that the two can be placed on one axis. Both are plotted as a percentage of the loss at the shared mean coordinate, which is what makes them comparable when the absolute loss scales are not. The horizontal axis walks the coordinate from that shared mean at $s = 0$ through the task’s own coordinate at $s = 1$ and out to twice that displacement.

5 STAGE C — DEMOGRAPHIC SCARCITY EXPERIMENT

5.1 C0 — Feasibility check (hard gate)

	Rows	Live images sampled
Archive 1	12,631 (76.2%)	0/30; domain fails to resolve
Archive 2	3,905 (23.6%)	30/30
Conditions above threshold, reachable subset		16
Of those, able to fund a 20-image support set		8
Conditions with the full dataset		83

Table 7: C0. Phototype grouping I–III against IV–VI, chosen from the observed counts.

Conclusion. The annotations are fully available and the images are not. Three quarters of the dataset sits behind a domain that no longer resolves, and recovering it depends on a release the dataset’s authors control. **The reachable subset supports eight usable conditions, which gives a five/two/one task split; one held-out test task cannot carry a paired interval.** It is therefore enough to build C1–C3 on, and not enough for the headline result of C4.

5.2 C1–C5 — Not run

	Item	Reason
C1	Domain module and splits	Not started. Buildable today on the reachable subset; not blocked.
C2	Frozen evaluation predictors	Not started; depends on C1.
C3	Optional sanity check	Not started; depends on C1 and C2.
C4	The scarcity sweep	Blocked by C0. The target-data equivalence statement the plan asks for needs a paired interval on held-out tasks, which eight conditions cannot provide.
C5	Downstream augmentation	Optional and conditional on C4.

Table 8: Stage C beyond the gate. The five items the plan places after C0, none of which was run, with the reason in each case.

6 STAGE D — FEW-SHOT FACE GENERATION

No new runs, as the plan directs. The existing negative result stands, reinterpreted as a task-family mismatch: the standard setup offers a handful of heterogeneous target domains rather than many instances of one reusable relation. A1 supports that reading, since the method’s advantage is concentrated where a reusable relation carries information a single target image does not.

7 DELIVERABLES

8 CONCLUSION

The criterion fixed before the stage began is met, and met narrowly: the correct source task together with the learned relation beats both relation-only transfer and target-only modelling when target data are scarce, with the margin over target-only modelling gone by twenty target images. The positive control confirms the mechanism is sound where the ground truth is known and locates the difficulty on images as shallowness rather than absence. Two further results are usefully negative: source evidence saturates below the smallest budget measured, so the method is cheap on the source side, and the non-linear transport map is not distinguishable from a linear one, so the architecture can be simplified.

#	Deliverable	Status
1	Gaussian-mixture mechanism and calibration	Complete
2	Source-dependence controls	Complete
3	Additive basis against the modulation arm	Not run — preconditions 0.1 and 0.3 unmet
4	Target-scarcity curve, crossing named	Complete; there is no crossing
5	Source-abundance curve	Complete; saturated from the first point
6	Transport-structure ablation	Complete
7	One demographic scarcity experiment	Not run — blocked by C0
8	Paired 95% intervals for every headline	Complete
9	Downstream augmentation (optional)	Not run — depends on C4
10	Few-shot face stress test (optional)	Not run — excluded by the plan

Table 9: The ten deliverables the plan fixes, and the standing of each at the close of this stage.

Three items would change what can be claimed. The unconstrained fine-tuning reference, which is implemented but was not run, would establish whether the remaining gap to the oracle belongs to the task or to the architecture. The branch integration of 0.1 unblocks deliverable 3 and is a repository operation. And the image release behind C0 is the only route to a task family in which a single target example does not already reveal the task — which, on the evidence of A1, is where this method has the most to offer.